HOW TO CHOOSE A TELESCOPE

The Ultimate Guide for Beginners



A FUN GUIDE TO UNDERSTANDING TELESCOPES AND THE COSMOS.

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SECTIONS

Front.....1

Sections.....you are here.....2

Quick Selection for Those in a Rush.....3

Where to Start.....6

Basics to Know.....7

Specifications on Telescopes.....9

Beginner Astrophotography.....14



My name is Isabella Adolf and I am the creator of this telescope guide. All the information in this document is from my personal experience as an amature astronomer at the Denver Astronomical Society and assosiate at Mile High Astronomy. I wanted to find a fun way to communicate science to the general public without making science seem too complicated or out of reach.

THE PURPOSE OF THIS GUIDE

Any one can do astronomy! It is one of the most accessible hobbies as everyone has access to the night sky! It can be overwhelming getting into astronomy with all of its own scientific terms and endless options of optics to choose from. This guide was made as a tool to share some knowledge about astronomy as well as an organized sheet to refer to when first learning about astronomy.

Science communication is a very important tool that can inspire people to learn and get involved with science. Because science can be hard to comprehend on its own, and because some sources are hard to sift through, I belive it is important to have outreach and documents like this to bridge the gap and help people understand important concepts.

NOTE: This is a simple guide to organize tons of information because it can be hard to retain it all when you are first starting out. Of course there are many more brands and extensive things to know about astronomy that has not been included in this document and is based on my personal expirence in the field.





FOR THOSE IN A HURRY, DON'T WORRY!

Beginner Visual	Pros	Cons	Applications
Moon FirstScope Telescope	• Perfect for looking at the moon.	• Can only really look at the moon. • Needs a table.	• Great as a first telescope for small children. It is lightweight but can only look at the moon.
70-80mm Travel Scopes	• Great first telescope. • Easy to travel with.	• Easy to grow out of quickly. • Not much detail in objects. • Horrible Tripod.	• Great as a compact portable first scope to look at the moon with great detail!
80mm Starsense Explorer LT	• Tripod step up from the 80mm travel scope. • Phone holder.	• Tripod is still not the sturdiest • Not much detail in objects	• Has the same visual quality as the travel scopes, but includes a better tripod which is worth the extra spending.
102mm Starsense Explorer	• More detail with objects. • Normalized sturdy tripod.	• Not made for looking at dimmer objects. • Very long.	• Increases detail and clarity on the objects viewable with an 70-80mm, and has an even sturdier tripod.
130mm Starsense Explorer	• Great for planets and dim objects. • Utilizes tripod instead of table.	• The eye piece is higher up, which is not ideal for small children.	• Increase in detail and amount of objects to look at. Able to look at dim objects like nebulas and clusters.
5" SCT Starsense Explorer	• Great for planets and dim objects. • Has more crisp and sharp details.	• Has a more set magnification comparatively due to focal length.	• Similar to the 130mm, but has a little more sharpness and clarity. Size and where the eyepiece sits may be easier for small children.
114-150mm Table Top Dobsonians	• Great for planets and dim objects. • Option for bigger aperture.	• Requires table or additional tripod fee. • More bulky shape.	• Same viewable objects as the 130mm, but the 150mm allows for way more detail. These require a table and can be harder to travel with.
8"-12" Dobsonians	• Amazing for planets, and dim objects including galaxies.	• Huge. • Heavy. • Hard to transport • May need chair.	• These allow you to see more deep space objects including galaxies! Can weigh from 45-65 lbs.

FOR THOSE IN A HURRY, DON'T WORRY!

Intermediate Visual	Pros	Cons	Applications
Equatorial Mounts	• More advanced at following the sky as the Earth rotates.	• Understanding of R.A and DEC. coordinates. • Often heavier. • Learning curve.	• These mounts naturally track the sky movement, and are nice because once an object has been found, you only need to use one slow motion control to follow it.
Computerized Go To Mounts	• Great for tracking objects. • Can find objects for you. • Astrophotography	• Heavier. • More expensive. • Learning curve with three star alignment.	• These mounts are made for tracking objects, great if multiple people are looking through the telescope. Also great for attaching cameras.
Beginner Astrophotography	Pros	Cons	Applications
Unistellar Odyssey	• Eyepiece feature. • Full- height tripod • Nikon Technology. • 80mm Aperture.	• Not the best imaging comparatively. • Expensive.	• Can be fun for outreach to see deep space objects, as it makes you feel like you're actually looking at the object.
ZWO Seestar Series	• Comes with everything you need. • Great software.	• Does not provide TIFF files. • Small aperture.	• Very affordable for beginner astrophotography, and has an intuitive software. • Includes EQ mode
Dwarf Lab Series	• Great Imaging • Similar to Seestar resolution	• Not the best software in comparison • No tripod included.	• Affordable and most comparable to the ZWO Seestars visually. • Provides TIFF files
Vespara Vaonis Series	• Bigger aperture. • TIFF files.	• No accessories included. • No EQ mode.	• Comparetively jumps in price, but has a 50mm aperture for more light, giving more detail or less overall capture time
Celestron Origin	• Biggest aperture. • TIFF files.	• Least transportable. • Expensive.	• Great reputation, huge aperture for amazing detail, well built software.

FOR THOSE IN A HURRY DON'T WORRY!

My Personal Favorites for Beginners

Beginner Visual: Celestron 5" SCT



Often when talking to beginners this telescopes seems to stand the most out. It is light and compact, only weighing 14 lbs. It is a Schmidt Cassegrain Telescope (SCT) which means it has both lenses and mirrors, and is much shorter than your regular telescope. The eye piece is lower comparatively to the other reflecting telescopes to compensate for the difference in size. This feature is often nicer for shorter children to look through the telescope. Because it is a SCT the visuals for objects are slightly more crisp and sharp compared to the other reflecters, but does have a more set magnification.

Beginner Astrophotography: Seestar S30 Pro



Seestar was made by ZWO which is a high quality and well known astrophotography company. The Seestar software is very intuitive and easy to use. There are many features that allow you to have as much or as little hands on for capturing and post processing. The seestar itself is very compact and light, great for transportation. The s30 Pro in particular has an amazing wide field of view for capturing huge objects, nigh scapes, and star trails. If you were to crop your image or zoom in, the results would be very similar to the resolution of the regular S30, making it a two in one!

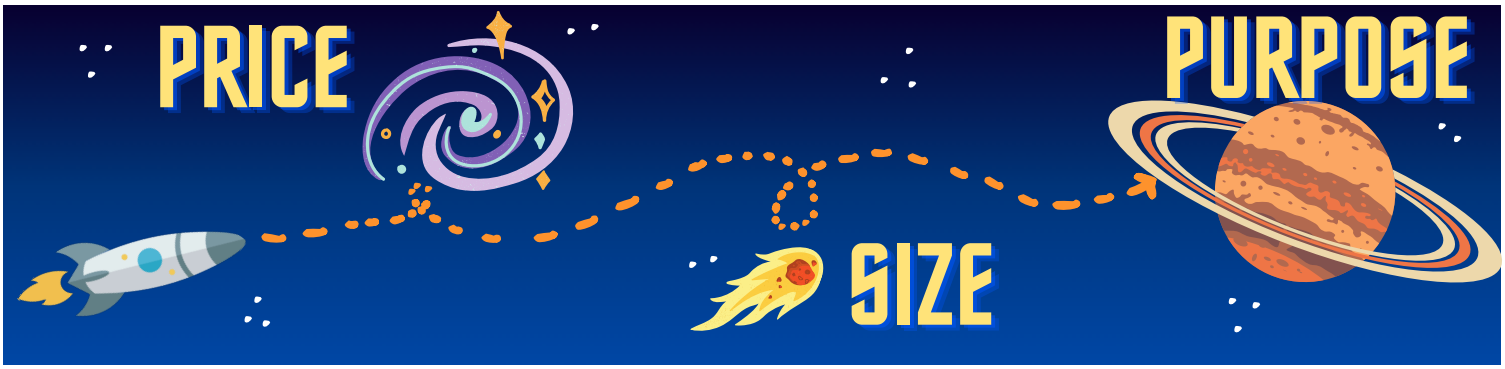
Beginner to Intermediate Visual and Astrophotography: Celestron 8" Evolution



This is the all in one telescope that you can grow with. This telescope has the feature to manually move the scope for quick, easy, and intuitive movement. It also has the feature of using the hand control to move the telescope including the GoTo feature to find and track objects. AND the mount has a higher weight capacity great for using cameras. The 8 inch SCT also have the feature where you can take out the secondary mirror and replace it with an astrophotography camera. This telescope has many amazing features to help you at each learning stage and has a big enough aperature to see amazing details!

WHERE TO START

There are three big things to consider when trying to figure out what kind of telescope to purchase, and they are also all connected. This is often the easiest place to start and can aid your decision making. They are Price, Size, and Purpose (in no particular order).



PRICE

Many people understand how much they are willing to spend on a telescope, so it is often the most easiest place to start. Beginner visual telescopes can range anywhere from $100 to $1000. Intermediate telescopes are often ranging from $900 to $4,000. Smart telescopes range from $300 to $4000, and serious astrophotography set ups can range from $5,000 to $10,000 and up. If you have a pretty large price range, Size and purpose can still steer you in the correct direction.

SIZE

The bigger in price often translates into the bigger in size of telescope. The size of the telescope often can be understood through aperture. Often times with a bigger aperature you also get a longer focal length. If you would like a bigger aperture but dont want a super long telescope, a Schmidt Cassegrain telescope may be the way to go. Along with bigger aperture, the more heavy your telescope gets which needs to be taken into consideration.

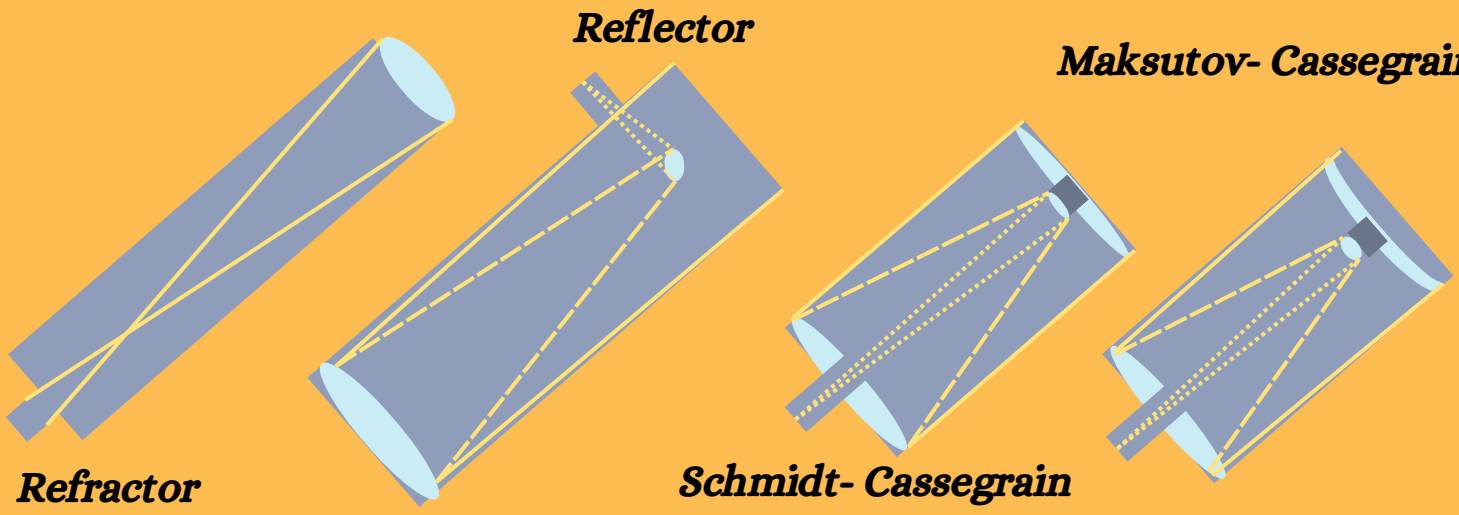
PURPOSE

Now Size affects your viewing. With the smaller telescopes you will see minimal details and the bigger you get with telescopes the more details and fainter objects you can see. This is where purpose is important; What are you wanting to see? Would you like to see mainly planets, stars, and fast moving objects, or would you like to see galaxies, star clusters, nebulas, and more dim objects? Are you interested in astrophotography and would you be willing to post process your own images?

BASICS TO KNOW

TYPES OF TELESCOPES

There are two basic telescopes types, refractors and reflectors. Refracting telescopes utilizes lenses to bend incoming light straight towards the eyepiece. Reflecting telescopes utilizes multiple mirrors to reflect light to the eyepiece. Now a little nicer types of telescopes are Schmidt-Cassegrain telescopes (SCT) and Maksutov telescopes. SCT have both features of a refracting and reflecting telescopes as it bends and folds light to have a huge focal length in a compact tube. These offer great details and higher magnification. Maksutov telescopes also refract and reflect light but have a thicker lens and curved secondary mirror to obtain more contrasting details specifically for planets.



TYPES OF MOUNTS

There are different types of mounts you can put your telescope on. Altitude/Azimuth is the most basic as they move up and down and left to right. Dobsonians too move like ALT/AZ but they sit on a table or the ground on a turn table like mount. Stepping it up there are equatorial mounts. EQ mounts are made to follow the motion of the sky and counter the Earths rotation and use right ascension and declination. These too are coordinates like ALT/AZ but are measured in time and degrees! Lastly there are computerized mounts. These mounts can only be moved with a hand controller but have the wonderful capability of finding objects and tracking them for you. By inserting the time, date, and location, then successfully doing a three star alignment where you center three distant and bright stars in the eyepiece, the telescope will know where it is and will give you a list of objects that are viewable that night! When choosing an object the telescope will instantly move to find your object and track it!

BASICS TO KNOW

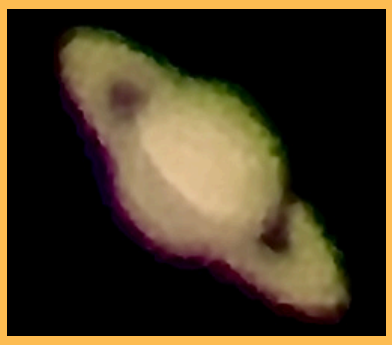
DETAIL VS. MAGNIFICATION

It is important to understand the difference between magnification and detail. Detail comes from how much light is let into the telescope. Bigger aperature telescopes are going to give more detail, and will expand the number of objects you can see, as many objects like galaxies and star clusters are dim. Magnification is provided from the eyepieces you use. They allow the ability to increase and decrease your field of view similar to zooming in and out. You will not be able to increse the detail you see by changing the magnification.

Eyepieces are counter intuitive. Bigger millimeter eyepieces will give you less magnification, and smaller millimeter eyepeices will give you higher magnification.



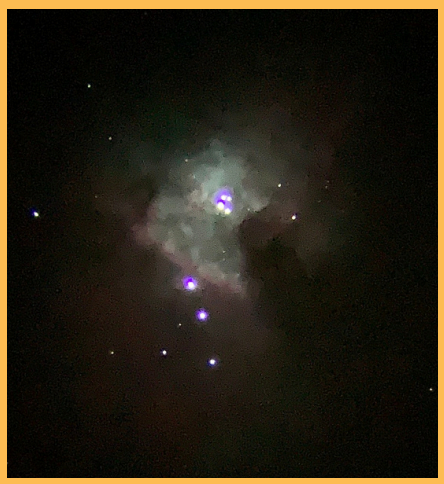
Bigger mm eyepiece used



Smaller mm eyepiece used

OUR EYES VS. WHATS ONLINE

Lastly, it is important to understand the difference between what our eyes can see and what cameras can see. Cameras are able to 'stare' and capture data way longer than our eyes can. They also have sensors that can apply color to what they are seeing, and they are able to take multiple images to stack and make a very detailed image. Unfortunately our eyes do not work like that but we are still able to see really cool details, just not the intense, colorful details we see in edited images.



Visual Astronomy



Astrophotography

BEGINNER TELESCOPES

Refractors   *Affordable, Small, but Easy to Grow Out of.*

70 mm to 102 mm Aperture

Price: Range from $100 to $400

Size: 1.5lbs to 14lbs

Viewing: Best for the moon, planets, and stars.

70/80 Travel Scope



70 mm to 80 mm Aperture

Price: $130, $150

Size: 2lbs, 4lbs

Viewing:



Because the aperature is so small we get minimal details in our objects. You wont be able to see the bands on Jupiter, and Saturn looks more like an oval shape.

80mm Starsense Explorer LT



80 mm Aperture

Price: $250

Size: 9lbs

Viewing:



This telescope will be the same viewing as the travelers scope. The biggest difference will be in the better tripod and the app to help guide your telescope.

102mm Starsense Explorer



102 mm Aperture

Price: $450

Size: 14lbs

Viewing:



This telescope upgrades the size which allows for more details, and the tripod which is bit more sturdy compared to the smaller telescopes.

BEGINNER TELESCOPES

Newtonian Reflectors

Entering Dim Object Zone.

114mm to 150mm Aperture

Note: Some of these are table top dobsonians with the option of a table top tripod.

Price: Range from $400 to $800

Size: 12lbs to 30lbs

Viewing: Great for the moon, planets and stars, introduces better viewing for dim objects like nebulas and star clusters.

114 mm Tabletop Dob.



114 mm Aperture

Price: $380-$530

Size: 13lbs

Viewing:



This scope gives a bit more detail and is available as the table top dobsonian. You will need a table for this item or purchase the additional tripod for $150

130 mm Starsense Explorer



130 mm Aperture

Price: $480-$500

Size: 19lbs

Viewing:



This comes in both a table top and tripod option. This is a great starter scope that gives just enough details on celestial objects.

150 mm Tabletop Dob.



150 mm Aperture

Price: $600-750

Size: 25lbs

Viewing:



This comes as the table top and can fit on the optional tripod. Is similar to the 130mm but is just a little bit bigger for more precise details.

BEGINNER TELESCOPES

Schmidt Cassegrain Telescope Reflectors

5 Inch Aperture

Introducing Crisp and Sharp Details

Price: $600

Size: 14lbs

Viewing: Great for the moon, dim objects like nebulas and star clusters. Best for sharp details for planets and stars comparitevly to the newtonian reflectors.

5" SCT Starsense Explorer



5 Inch Aperture

Price: $650

Size: 14lbs

Viewing:



Similar aperture to the 130mm, but this is a SCT so it will provide a bit more clarity and sharpness with planets comparatively. This is also lighter and sometimes easier for smaller children to look through.

Newtonian Reflector Dobsonians

8 to 12 Inch Aperture

Gives the best visual observing at the cost of weight

Price: $800-$1600

Size: 45 lbs to 85 lbs

Viewing: Great for all objects especially really dim objects like galaxies!

Starsense Explorer Dobsonians

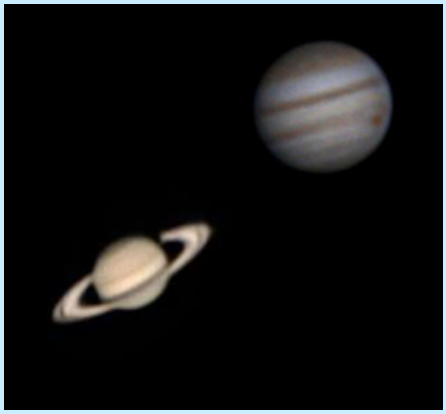


8 inch
Price: $850
Size: 45 lbs

10 inch
Price: $1200
Size: 55 lbs

12 inch
Price: $1600
Size: 85 lbs

Viewing:



This is a huge step up from the previous telescopes. They are only on the Dobsonian mount and do weigh a ton more, but the increase in detail is through the roof! You can distinguish between the rings on Saturn! You are also able to see more details in galaxies! The main downside could be the length which may push you to look at the SCTs.

INTERMEDIATE TELESCOPES

Telescopes on Beginner Equatorial Mounts

Omni CG-4 Mount **Price: $400** **Size: 45 lbs**



To step up your set up, you may be interested in an equatorial mount. Unlike the previous options that move up and down, left and right (ALT/AZ), an equatorial moves in an arc motion using right ascension and declination to smoothly follow objects in the sky. These mounts are usually more heavier and bulky than a basic mount, but offer more simplicity to follow objects. These mounts can also hold heavier telescopes as they have counterweights to balance your whole rig. Equatorial mounts also can come computerized which are amazing for astrophotography!

Telescopes on Computerized Go To Mounts

SE Mount



Price: $650-$1,700
Size: 16 lbs-33 lbs

Evolution Mount



Price: $1,800-$3,000
Size: 35 lbs-61 lbs

CPC Mount



Price: $3,000-$4,500
Size: 61 lbs-100 lbs

There are many types of computerized go to mounts that all work in a similar fashion. Each have a hand controller to electronically move the telescope where you want it to point. To use the GO TO feature you need to insert the date, time, and location and do a three star alignment so that the scope knows where it is and what the sky looks like. The three star alignment is usually where folks get stuck on as you have to locate and center three bright stars quite a distance apart, in the telescope. Once done successfully, the telescope will give a list of objects for that night that you can choose from and will automatically find and track the object you choose! Each of these three mounts work the same, the main difference is that the mounts get progressively more heavy duty for bigger telescopes. The Evolution and CPC mount have the capability of being moved manually and can hold more weight, great for adding on cameras!

INTERMEDIATE TELESCOPES

Telescopes on Computerized Go To Mounts Continued

Maksutov-Cassegrain Telescopes Price: $750-$5,000 Size: 20 lbs-100 lbs



Maksutov-Cassegrain telescopes are very nice quality scopes that will bring higher contrast and sharpness in details of the objects you are observing. They do come only in limited amounts of sizes around 4 inches to 7 inches as the glass making for the corrector plate and curved mirror can be difficult to make. If you have the funds and the time, there are some small businesses out there that dedicate their skills to making bigger high quality maksutov-cassegrain telescopes!

Schmidt-Cassegrain Telescopes Price: $1,000-$5,000 Size: 20 lbs-100 lbs



Schmidt-Cassegrain telescopes are an all time favorite. They range from 5 inches to 16 inches and also give high quality imaging similar to the Maksutov-Cassegrain telescopes. The glass for SCTs are a bit easier to make comparatively to the Maksutov-Cassegrain telescopes, making bigger sizes easier to find. In comparison to newtonian reflectors, SCTs are more compact. They also have more set in place optics so you rarely if not at all have to collimate them. SCTs also have a higher base in magnification and narrower fields of view.

Edge HD Telescopes Price: $2,000-$9,000 Size: 50 lbs-150 lbs



Edge HD telescopes are a special type of Schmidt-Cassegrain telescope. They often have built in correctors to fix off-axis coma and field curvature. This gives you more sharp and flat field stars all the way to the edge of your field of view. Edge HD telescopes have amazing optics for viewing and are great if you have a particular prefference for perfect stars. Often times Edge HD telescopes are used for astrophotography because of their flat field aspect.

BEGINNER ASTROPHOTOGRAPHY TELESCOPES

Main Big Brands



Zwo Seestar



DwarfLab



Vaonis



Celestron Origin

Things to Know

Smart Telescopes Are Not for Visual

You can not visually look through a smart telescope. They only connect to an app on your phone where you can see the accumulation of photos compile of your objects.

Can Be Very Hands Off

They can do the whole astrophotography process for you including finding your object, taking exposures, stacking images, and editing them as well. They also give you the option to post process your own images

Not Made for Planetary Viewing

Smart telescopes are made for dim, deep space objects. Although planets are super close to Earth, they are very tiny comparatively to farther away but bigger objects. You can still try to capture planets but the images will not have much detail.

The image to the right shows the angular size comparisons of some planets and deep space objects to the moon. Notice how small planets are angularly to the moon.

